# COMPUTER VISION PROJECT SUMMARY

## Face Recognition: From Handcrafted Features to Deep Embeddings

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## PROJECT DETAILS

- ***Role*** → Computer Vision & Machine Learning Engineer (group of four)
- ***Setting*** → KU Leuven, course H02A5a Computer Vision, Group assignment
- ***Technical Stack*** →
  - Python: NumPy, Pandas, scikit-learn, scikit-image, OpenCV
  - Face detection: HAAR cascade, YuNet
  - Deep embeddings: DeepFace (FaceNet512, ArcFace, VGG-Face)
  - Visualization: Matplotlib, t-SNE
- ***Core Domains*** →
  - Face detection and image preprocessing
  - Handcrafted features (HOG, SIFT + VLAD) and PCA (eigenfaces)
  - Deep face embeddings used as frozen feature extractors
  - Small-data classification (KNN, SVM, Random Forest, voting ensemble)
  - Cross-validation without data leakage, ablation studies

## PROBLEM STATEMENT

The task is to recognise two people, **Jesse** (label 1) and **Mila** (label 2), in photos. Every other face, including **lookalikes**, must be labelled 0. The setup is close to a real-world case: only **79 training images** (30 Jesse, 29 Mila, 20 lookalikes), but a test set of **1,816 images**. Three challenges shaped the project:

- **Very little data:** with only 79 images, large feature vectors easily lead to overfitting.
- **Lookalikes and bystanders:** the model must reject faces that are similar to the targets, or that belong to other people in the photo.
- **Several faces per image:** the system must first find the right face before it can classify it.

### The objective

🎯 ***Main goal*** → *build a complete face recognition pipeline, from detection to classification, and compare handcrafted features with deep embeddings.*

*Steps required:*

***First objective*** → *detect and crop faces reliably, even when they are tilted or when several people appear.*

***Second objective*** → *compare six feature sets and four classifier families with a fair 5-fold cross-validation.*

***Third objective*** → *run ablation studies and publish predictions for the test set with the best setup.*

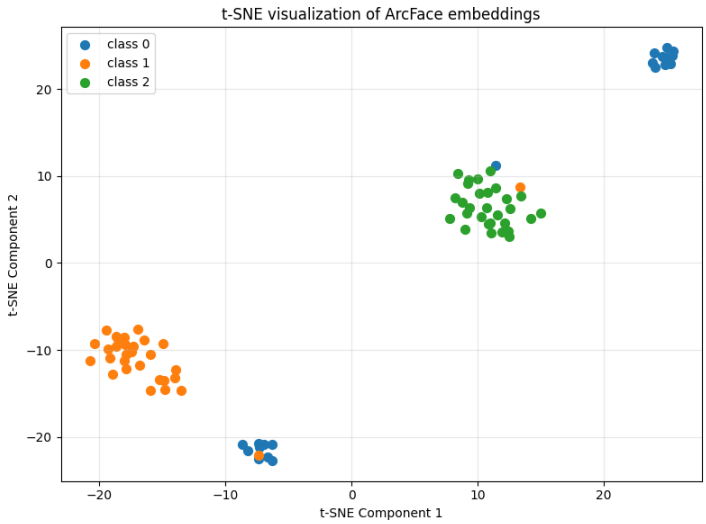
## APPROACH

- **Face detection:** a HAAR cascade was the baseline. YuNet, a small CNN detector, became the main detector because it gives tighter crops and handles tilted faces better. For training images with several faces, the face closest to its class average is kept. At test time every face is classified, and it is accepted only if the confidence is above 0.60.

- **Features:** HOG (tuned with t-SNE), SIFT + VLAD, PCA eigenfaces (10,000 values reduced to 36), and three deep embeddings (FaceNet512, ArcFace, VGG-Face) used without fine-tuning.
- **Classifiers:** KNN, SVM (RBF, linear, polynomial), Random Forest and a soft voting ensemble.
- **Evaluation:** 5-fold stratified cross-validation with macro F1, so all three classes count equally. PCA and VLAD were fitted on the training folds only, to avoid leaking information.
- **Ablations:** PCA before the classifier, data augmentation, and an extra "other people" class made of 30 faces from the LFW dataset.

## RESULTS

| Feature set | Best macro F1 (5-fold CV) | Best classifier |
|---|---|---|
| HOG | 0.776 ± 0.125 | Linear SVM. Best handcrafted feature, but unstable. |
| SIFT + VLAD | 0.776 ± 0.073 | Voting ensemble. Weak on most classifiers, as it loses face layout. |
| PCA (eigenfaces) | 0.735 ± 0.129 | Random Forest. Compact, but not discriminative enough. |
| FaceNet512 | 0.973 ± 0.033 | Linear SVM. Best overall. |
| ArcFace | 0.960 ± 0.034 | Stable across almost all classifiers. |
| VGG-Face | 0.963 ± 0.050 | Voting ensemble. Likely inflated: the model was trained on data that overlaps with this dataset. |




*t-SNE of ArcFace embeddings: the three classes form clear groups (class 0 = lookalikes, 1 = Jesse, 2 = Mila).*

- **Deep embeddings win:** FaceNet and ArcFace reach about 0.96–0.97 macro F1, against about 0.78 for the best handcrafted features.
- **Lookalikes are the hardest class:** with HOG, Jesse is easiest (F1 0.92), then Mila (0.82), then lookalikes (0.72), which matches what t-SNE showed.
- **PCA helps big features:** it raised HOG with SVM from 0.75 to 0.84, but gave almost no gain on compact embeddings.
- **Augmentation hurt:** augmented points sit too close to the originals in embedding space, so they add little new information.

- **The “other” class lowered CV F1** (0.96 to 0.94), but it was kept in the final model because CV cannot measure its real purpose: rejecting bystanders in the test photos.

## CONCLUSIONS

The final model uses FaceNet embeddings with a soft voting ensemble (KNN, SVM, Random Forest) and the extra “other” class. It classified 2,735 faces across the 1,816 test images. The main lessons were that pre-trained deep embeddings beat handcrafted features by a wide margin on tiny datasets, that preprocessing (detection and face selection) matters as much as the classifier, and that cross-validation must be set up carefully to avoid leakage.